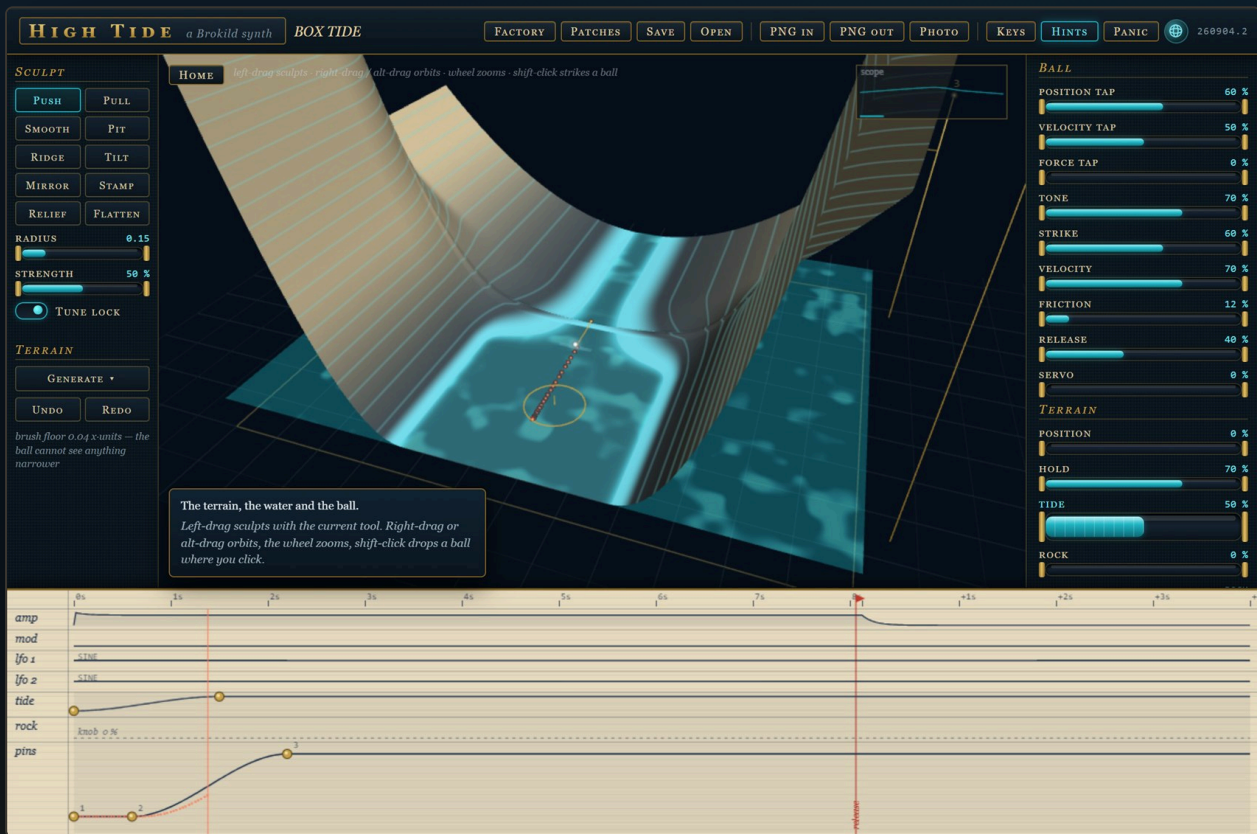


# HIGH TIDE

*A wavetable synth that stores no waveforms. A frame is a bowl on a sculpted terrain, the waveform is what a mass does when it is dropped into it, and position is where the ball is. The tide floods the passes; pins on a timeline tow the ball where you want it. Seventeen starters to play at once, and every control tells you what it is for.*



## HIGH TIDE

terrain · ball · tide · pins · rock · 17 starters · 12 terrains · world rack

## BROKILD

free · VST3 + standalone · 64-bit

Every wavetable synth rests on one unquestioned axiom: a frame is one cycle of waveform data, read by a phasor at the pitch. Position, morphing, mip-maps and the stack of frames all follow from it. High Tide keeps the picture and refuses the axiom.

**A frame is a bowl.** Each column of the terrain is a curve of potential energy. The sound is the trajectory of a mass dropped into that curve, integrated by Newton's law at audio rate. Nothing stores a waveform anywhere in the instrument. A parabola gives a sine, a flat floor with walls gives a triangle in position and a square in velocity, a narrow pit in the floor gives a pulse each time the ball dives through it. You author the cause. The instrument plays the consequence.

**The reader is a body, not a phasor.** It has energy, so a hard strike climbs higher up the walls and brightens. It has friction, so a note decays by sinking to the floor, where every bowl is locally a parabola, and every note therefore ends as a sine. It has momentum, so nothing snaps.

**Position is a place, not a number.** The frames stand along a second axis, and the ball moves along it slowly, towed by a tether toward a target.

Between two valleys there is a ridge. A low tide leaves the ridge standing and the ball must be dragged over it; a high tide floods the pass and the ball strolls across. Morphing is travel, and the transition between two frames is the shape of the pass, which you can carve.

## Pitch is exact by construction

A textbook result says the period of a bowl depends only on its width at each height. Any bowl whose width matches the reference parabola's is exactly isochronous: far from a sine, and in tune whatever the strike. TUNE LOCK in the sculpt rail keeps that rule while you carve. Turn it off and the departure paints the bowl red: that band is the pitch bend the player will hear. The bench measures a tune-locked bowl within 0.01 cents across a 40 dB range of strike.

**Nothing to erase.** The reference bowl plays a sine with every harmonic below  $-119$  dB, at 110.000 Hz. There is no buffer of samples in the voice for that number to come from. It comes from the physics.

Open the plug-in, or the standalone. It opens on **SOFT KEYS**, the first of the starters: a warm poly key whose timbre travels a little while you hold it.

**Two things to do first.** Hover anything at all — a knob, a button, a lane of the timeline — and after a moment a note appears beside it saying what it is and how it is meant to be used. And open **FACTORY**: the **STARTERS** group is seventeen ordinary instruments (basses, leads, keys, organs, pads) that behave the way you expect, and the **TERRAINS** group below it is the instrument showing what else it can do. The **HINTS** button in the header turns the notes off once you know your way around.

**Play a note.** The pearl drops into the valley and its swing is drawn as a streak on the terrain. Watch it travel along the position axis as the pin tows it. Hold the note and it stays where the last pin left it; let go and the release pin takes it back.

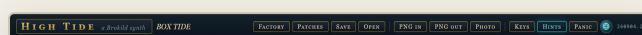
**Move TIDE.** The water rises on the relief and the flooded columns rise with it so their floors sit on the water line. That is exactly what the ball feels: a ridge under water is not there. Low tide, and the ridge stands.

**Sculpt.** Pick PUSH and drag on the terrain. With TUNE LOCK on, the other wall follows and the note

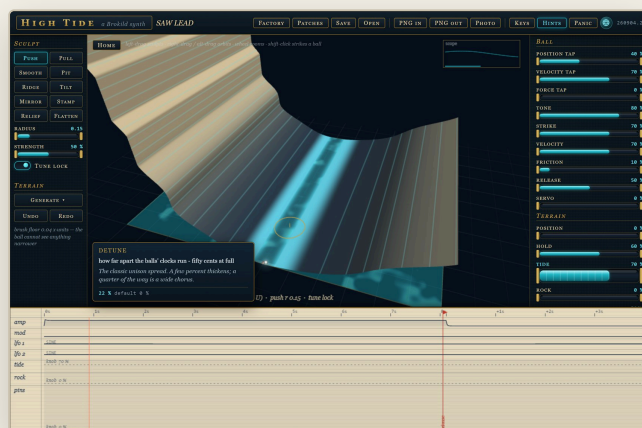
does not move. Every brush stroke while a note sounds throws the ball, so every touch is audible.

**Pin.** Click in the bottom lane of the timeline to add a pin; drag it; right-click it to remove it; double-click it to change its ease. Points left of the red flag count from the key going down; points right of it count from the key coming up. Shift-drag a loop.

**Rock.** Raise ROCK on a double-well terrain (CASCADE, GROUNDSWELL) and listen to the octave drop in.



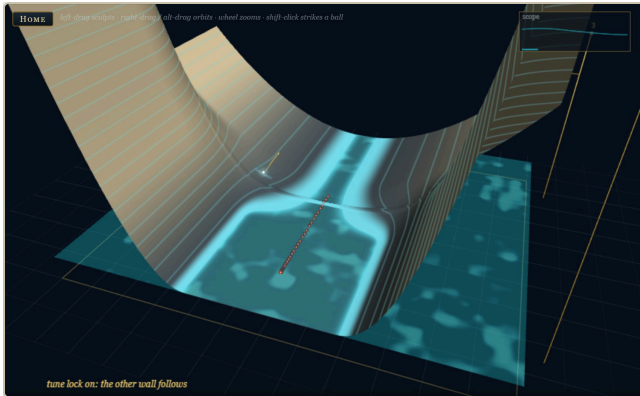
FACTORY · PATCHES · SAVE · OPEN · PNG IN · PNG OUT ·  
PHOTO · KEYS · HINTS · PANIC · the world rack.



Every control says what it is, how it is used, and what it is set to against its default. The note is always placed beside the control, never over it.

The terrain is a heightmap of 512 samples across the swing axis by 128 columns along the position axis, in height units where the reference bowl is  $\frac{1}{2}x^2$  and a full-scale strike reaches  $x = \pm 1$ . Every feature has one fixed meaning to the ear, and a sculptor learns the vocabulary once.

| FEATURE                         | SOUND                                                                            |
|---------------------------------|----------------------------------------------------------------------------------|
| floor of a valley               | the note and its quiet timbre; its curvature is the pitch; every decay ends here |
| walls                           | the loud timbre — steep is bright and square-ish, gentle is round                |
| a shelf halfway up              | a timbre that exists only at mezzo-forte                                         |
| a pit in the floor              | a pulse: the ball dives through and comes out with a spike in its velocity       |
| two pits with a hump            | an octave regime that opens when the strike clears the hump                      |
| asymmetry                       | even harmonics; mirror the bowl and they vanish                                  |
| a ridge along the position axis | a morph threshold; its height is the force needed to cross                       |
| the pass over a ridge           | the sound of the transition — wide and it strolls, a notch and it snaps          |
| a tilt along the position axis  | the direction timbre drifts when the playing gets loud                           |
| roughness                       | grain, down to the brush floor                                                   |



The terrain view. Contour lines every tenth of a unit, the water at the tide line, the pearl and its streak, the tether, the pins as stakes along the far edge.

## The centre line, and why a bowl can be rich and still in tune

Write a bowl as a centre line with the parabola's width around it,  $x = c(h) \pm \sqrt{2h}$ . The width at every height is then the reference parabola's whatever  $c$  does — so the bowl's centre line may wander with height however it likes and **the pitch does not move**. A straight centre line leans the bowl and brings in the even harmonics; a wavy one gives a wave nothing like a sine, exactly in tune. That construction is what every starter is built from,

and it is what TUNE LOCK maintains while you sculpt.

## Three taps

The same motion can be heard three ways.

**POSITION TAP** is where the ball is: the round sound. **VELOCITY TAP** is how fast it is going: brighter, and where a pit's pulse lives. **FORCE TAP** is the wall's push on it: the hardest of the three, and it hears a wiggly wall even while the position stays a clean swing. Mix them. **TONE** is a radiation lowpass after the taps, from 200 Hz to 20 kHz.

## The brush floor

The ball cannot see a feature narrower than the distance it travels in one step, and a feature it half-sees is noise. So the brush refuses a radius under 0.04 units, and a fast ball is handed the terrain blurred to about its own step: a high note sees a smoother world, the way a wavetable at a high pitch sees fewer harmonics. The engine also subdivides its step where the terrain is stiff, so a wall bounce is resolved rather than guessed.

Sculpting happens on the terrain itself. Left-drag paints with the current tool; the brush footprint is a ring under the pointer. Right-drag or alt-drag orbits the camera, the wheel zooms, HOME resets the view. Shift-click drops a ball where you click.

| TOOL        | WHAT IT DOES                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------|
| PUSH / PULL | raise or lower the terrain under a raised-cosine brush; RADIUS and STRENGTH below                      |
| SMOOTH      | relax the terrain under the brush toward its neighbours                                                |
| PIT         | dig a narrow pit — a pulse in the velocity tap                                                         |
| RIDGE       | raise a whole band of columns: a morph threshold                                                       |
| TILT        | slope the band of columns along the position axis                                                      |
| MIRROR      | mirror the column about its floor: the even harmonics leave                                            |
| STAMP       | drop a known bowl into the columns under the brush: SINE, BOX, PIT BOWL, DOUBLE WELL, SHELF            |
| RELIEF      | edit only the floor height along the position axis — the passes — on the strip at the foot of the view |
| FLATTEN     | put the reference bowl back under the brush                                                            |



## SCULPT

PUSH

PULL

SMOOTH

PIT

RIDGE

TILT

MIRROR

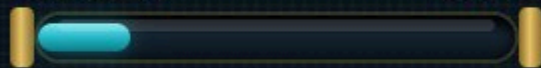
STAMP

RELIEF

FLATTEN

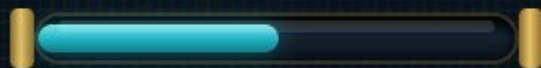
RADIUS

0.15



STRENGTH

50 %



TUNE LOCK

## TERRAIN

GENERATE ▼

UNDO

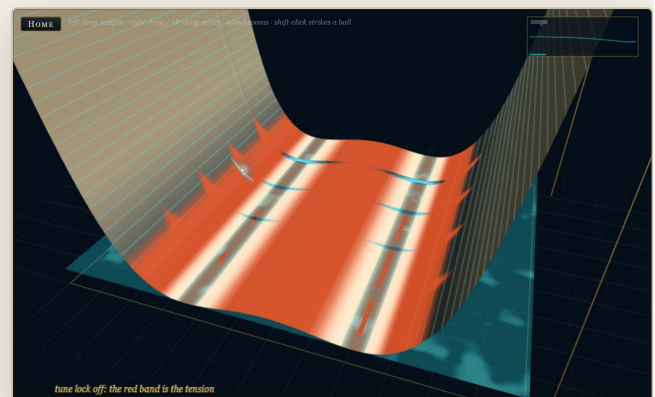
REDO

*brush floor 0.04 x-units — the ball cannot see anything narrower*

The sculpt rail: tools, brush radius and strength, TUNE LOCK, GENERATE, UNDO and REDO.

### Tune lock

With TUNE LOCK on, whatever you do to one wall of a bowl, the other wall is rebuilt so that at every height the width equals the reference parabola's. The pitch cannot move; the shape can do anything. A column with a pit or a shelf is not monotone and the lock skips it, and says so. With the lock off, every column is coloured by how far its width departs from the rule: that red is tension, the pitch bend a strike will produce. It is not a knob; it is something you can see you have carved.



CASCADE with TUNE LOCK off: a double well is nowhere near the width rule, and the whole terrain says so.

### Generate, undo, and where terrains come from

GENERATE offers four starting grounds: six valleys with ridges between, one double well with rising ridges, the reference bowl everywhere, and valleys with a little grain. Ctrl-Z and Ctrl-Y undo and redo both terrain and pins. PNG OUT writes the terrain as a 16-bit greyscale PNG and PNG IN reads one back, so any image editor is a terrain editor; PHOTO reads any picture and uses each row's brightness as a bowl profile, the way Photo Synth does.

## CONTROL MEANING

|                 |                                                                                       |
|-----------------|---------------------------------------------------------------------------------------|
| <b>STRIKE</b>   | the energy of a full-velocity hit; at 100 % the ball reaches the walls at $x = \pm 1$ |
| <b>VELOCITY</b> | how much key velocity scales the strike                                               |
| <b>FRICTION</b> | energy lost while the key is down; zero is frictionless and the note holds forever    |
| <b>RELEASE</b>  | energy lost once the key is up: the ball sinking to the floor                         |
| <b>SERVO</b>    | zero trusts the bowl's own period; one measures it and corrects the clock each cycle  |
| <b>TONE</b>     | the radiation lowpass                                                                 |

The amplitude envelope in the AMP section is an ordinary output VCA, kept for the sake of every DAW. The ball's own energy is the other dynamic: a note under FRICTION 0 and AMP SUSTAIN 100 % rings as long as the key is down, and its timbre closes as the ball sinks. The two are lanes on the timeline, side by side.

*BALL*

|                     |      |
|---------------------|------|
| <b>POSITION TAP</b> | 40 % |
| <b>VELOCITY TAP</b> | 70 % |
| <b>FORCE TAP</b>    | 0 %  |
| <b>TONE</b>         | 80 % |
| <b>STRIKE</b>       | 70 % |
| <b>VELOCITY</b>     | 70 % |
| <b>FRICTION</b>     | 10 % |
| <b>RELEASE</b>      | 50 % |
| <b>SERVO</b>        | 0 %  |

*TERRAIN*

|                 |      |
|-----------------|------|
| <b>POSITION</b> | 0 %  |
| <b>HOLD</b>     | 60 % |
| <b>TIDE</b>     | 70 % |
| <b>ROCK</b>     | 0 %  |

The right rail: BALL, TERRAIN, VOICE, AMP, MOD, LFO 1, LFO 2. Every control is a brass-ferruled tide gauge.

**Servo and tension**

A bowl that breaks the width rule bends its pitch with the strike. A box bowl, for instance, plays a soft note a long way below a hard one, because a ball crawling across a flat floor takes its time. That



is a sound, and a good one on a bass. When it is not wanted, SERVO measures the period the bowl actually produced and eases the clock onto the note, one cycle late. The bench holds a box within a few cents at SERVO 70 %.

## **Unison**

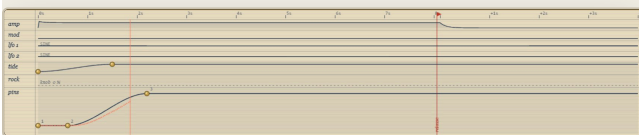
UNISON puts up to four balls in the same bowl.

**DETUNE** runs their clocks a little apart — fifty cents

across the fan at full — which is the familiar unison spread, and the ball that plays your note is never one of the detuned ones, so the pitch stays exact however wide it gets. SPREAD scatters how hard each ball is struck instead, so they differ in brightness and phase; in an isochronous bowl that is almost no detune, and in a box it is a great deal, because in a box energy is pitch. WIDTH stands them across the stereo field.

The bottom of the window is a timeline that runs from the moment a key goes down, per voice. Each lane draws the real curve of a modulator against time: AMP and MOD envelopes as their shapes, LFO 1 and LFO 2 as their waveforms at their rates. Below them three point lanes, TIDE, ROCK and PINS, are yours to draw on.

**A pin is a target, not a coordinate.** The pin lane defines where the ball is towed at every moment; HOLD is the strength of the tether; the terrain decides how the ball gets there. A ridge in the way slows it or stops it. TIDE floods the relief: everything below the water line is a free sweep, everything above it is a wall. High tide and high hold is the video editor, the ball goes exactly where the pins say. Low tide and low hold is the pure landscape. Everything between is a terrain that partly has its way.



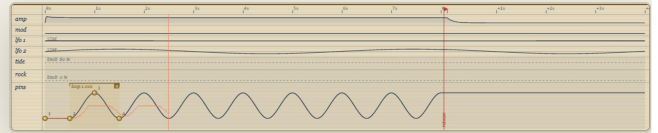
The timeline: envelopes and LFOs drawn as they are, then TIDE, ROCK and PINS. The red flag is the release marker; to its right, time counts from the key coming up.

## Two anchors and a loop

Points left of the release marker count from note-on. Points right of it count from note-off, and the release curve starts from wherever the lane was when the key came up, so there is never a jump. A held note stretches the middle, exactly as the envelope does. Shift-drag on a point lane to set a loop: while the key is down, the points inside it cycle. That is how a hand-drawn LFO along the terrain is made. Click the small box at the loop's end to clear it.

Each point carries an ease into it: HOLD steps at the point's time, LINEAR goes straight, SMOOTH eases in and out. Double-click a point to cycle

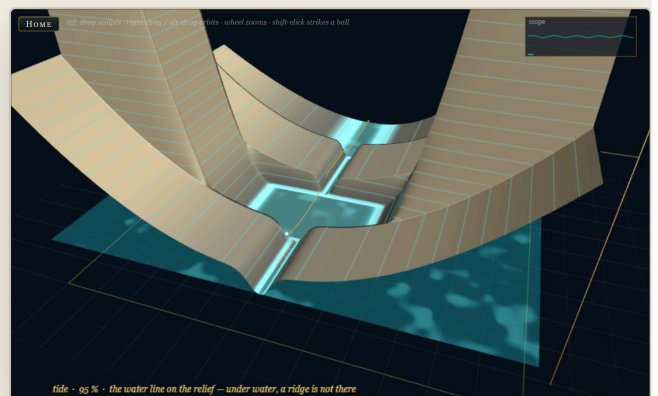
them. An empty TIDE or ROCK lane means the knob rules; a lane with points owns the value for the whole note and the knob is shown as a faint line for reference.



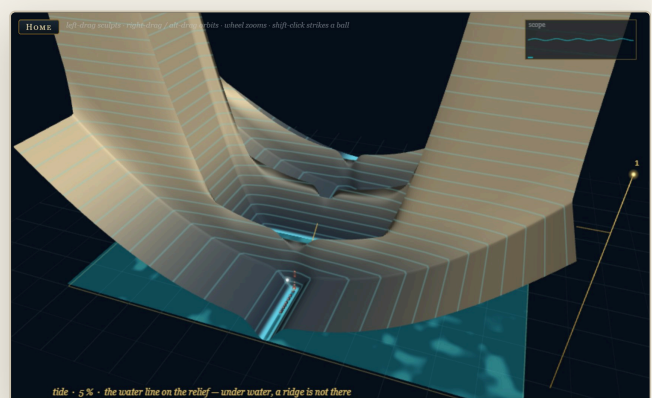
SPRING TIDE: a loop from half a second to a second and a half, cycling the ball between two valleys while the key is held. The coral dots are the wake of the last note.

## The wake

The last note's actual path is drawn over the pin lane in coral dots and as a trail on the terrain. Where it follows the pins the tide carried; where it lags or stops, the terrain won. The same key at the same velocity leaves the same wake, to the sample.



High tide: the passes are flooded and the ball is towed anywhere.



Low tide on the same terrain: the ridges stand and the ball is stranded in its valley.

A ball on a still terrain can never be chaotic; that is a theorem, and it is what makes everything above playable. Add one ingredient and the whole bifurcation literature opens. **ROCK** tilts the whole terrain like a see-saw, at a ratio of the note.

**BREATH** steepens and relaxes the bowls instead. Both are the same knob, **ROCK**, under **ROCK MODE**.

In a double well as **ROCK** rises: the ball wobbles in one pit, following the drive. Then it takes two rocks to complete one figure, and the pitch drops an exact octave and goes hollow. Then four, then eight, then the cascade closes and the ball hops between the pits at times nobody chose. Inside the chaos there are windows where it settles into a three-rock figure. This is not white noise: the drive stays as a strong line and the disorder is in how the ball disagrees with it. The bench finds the sub-octave at **ROCK 30 %** on the **CASCADE** terrain, and proves the chaos as chaos: two strikes a millionth apart diverge to order one within three seconds.

The rock rate is the note, so the pitch survives the chaos. **ROCK RATIO** offers 1/1, 2/1, 1/2, 1/3, 3/2

and 3/1, or **FREE** with **ROCK RATE** in hertz. A rock at the note also pumps an isochronous bowl until friction balances it, so a rocked note sustains like a bowed string and decays down the bowl when the rocking stops. **BREATH** at 2/1 is the parametric pump, a child on a swing; it has finite power, so it settles at a level rather than growing without end.

**Rock is on the timeline too.** A **ROCK** lane that rises over a note walks the sound from a clean tone into period doubling and out the other side, every time the key is pressed, the same way. **CASCADE** does exactly that over four seconds.

## Voices

**VOICES** is **POLY** (twelve), **MONO** (every key restrikes and restarts the timeline, the pitch glides over **GLIDE**) or **LEGATO** (an overlapping key only moves the pitch and the timeline runs on). A released key in **MONO** falls back to the key still held without a new strike. **TUNE** is twelve semitones either way; the pitch wheel bends two. The mod wheel and channel pressure lift the **TIDE**.

## The mod envelope and two LFOs

MOD is a second ADSR with a TARGET and a bipolar AMOUNT. Each LFO has a RATE, a SYNC to the host clock (FREE, 4/1 down to 1/16 with triplets), a SHAPE (SINE, TRI, SAW, SQUARE, S&H) and the same TARGET list: POSITION pushes the ball's target along the terrain, TIDE, ROCK, HOLD and TONE offset those controls, PITCH is vibrato. Free LFOs restart at every key so a note is the same note twice; synced ones ride the transport.

## Patches

FACTORY lists the twelve built-in terrains. SAVE writes a patch as a JSON file into

`Documents\Brokild patches\High Tide\`, with the terrain inside it as a 16-bit PNG, the pins, every control and the rack. OPEN reads one; PATCHES lists the folder, subfolders as groups. A patch is the whole instrument: terrain, pins, controls and rack together.

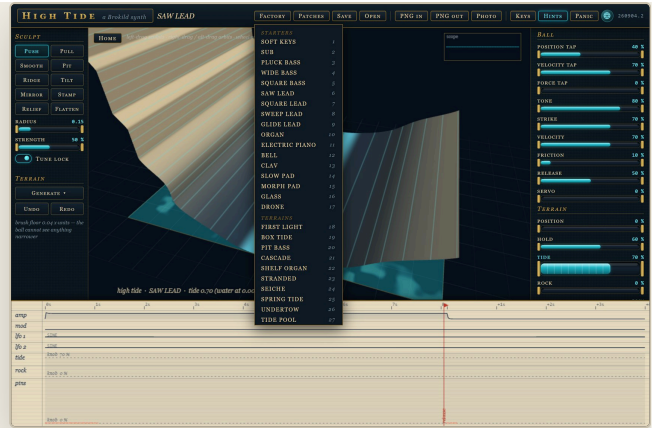
## STARTERS — the safe ground

Seventeen ordinary instruments, every one of them built on tune-locked bowls with no drive, so the note is the note whatever the strike. They are the place to begin, and the place to start editing from.

| STARTER        | WHAT IT IS                                                       |
|----------------|------------------------------------------------------------------|
| SOFT KEYS      | a warm poly key whose timbre travels a little while held         |
| SUB            | a mono sub on the position tap, dark and short                   |
| PLUCK BASS     | heavy friction: the brightness closes as the ball sinks          |
| WIDE BASS      | two detuned balls, wide                                          |
| SQUARE BASS    | a box held in tune by the servo                                  |
| SAW LEAD       | a wavy centre line on the velocity tap, two balls detuned        |
| SQUARE LEAD    | a box with the servo on, bright                                  |
| SWEEP LEAD     | a pin sweeps the centre line open: a filter sweep with no filter |
| GLIDE LEAD     | legato with portamento, a slow breath on the tide                |
| ORGAN          | frictionless and sustaining, two balls barely apart              |
| ELECTRIC PIANO | velocity into brightness, the bell decaying into the body        |
| BELL           | a pit bowl struck hard, long tail                                |
| CLAV           | the force tap, short and hard                                    |
| SLOW PAD       | four balls, slow attack, the tide breathing                      |
| MORPH PAD      | a pin strolling across three centre lines over eight seconds     |
| GLASS          | a wavy bowl, wide open tone, very long release                   |
| DRONE          | frictionless, four balls, the position wandering by itself       |

## TERRAINS – the instrument showing what it is

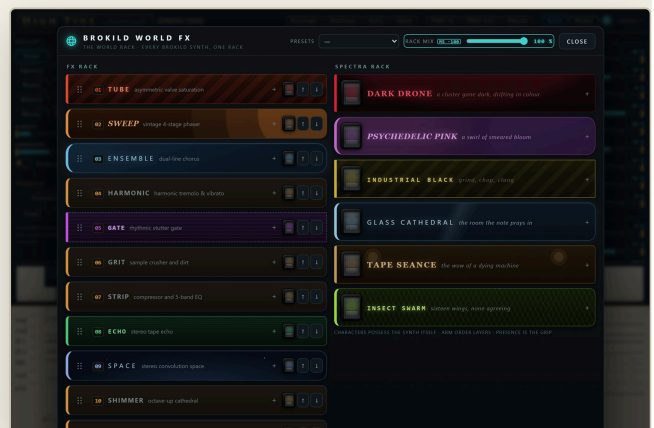
| TERRAIN     | WHAT IT IS                                                                   |
|-------------|------------------------------------------------------------------------------|
| FIRST LIGHT | two sine valleys, a small ridge, a slow sweep and a release pin              |
| BOX TIDE    | a box that morphs into a sine over a high ridge while the tide rises         |
| PIT BASS    | mono, glide, a pit into a double well, rock at 1/2                           |
| CASCADE     | one double well; the pin sweep and the rock lane rise together               |
| SHELF ORGAN | shelved bowls, three frictionless balls, sustain                             |
| STRANDED    | four valleys behind tall ridges at low tide; the LFO pushes and mostly fails |
| SEICHE      | BREATH at 2/1 bowing a box: the parametric pump                              |
| SPRING TIDE | a looped pin lane between two valleys, the LFO on the tide                   |
| UNDERTOW    | mono bass on the force tap with a tilted box                                 |
| TIDE POOL   | a grained sine with the servo holding pitch                                  |
| NACRE       | four balls in a sine bowl, wide                                              |
| GROUNDSWELL | legato sub-bass, a double well rocked at 1/3                                 |



FACTORY, in two groups: the starters first, the terrains below.

## The world rack

The teal globe in the header opens the Brokild World FX rack: twelve pedals after the voice sum, the SPECTRA characters that reach inside the voices (their detune is fanned across the balls, their sag is keyed to the gate, their tremolo and filter ride the taps), and five MACROS that are the rack's only host parameters. A fresh rack is bit-identical to no rack at all.



The Brokild World FX rack, shared by every Brokild synth.



KEYS shows two octaves; ZSXDCVBHNMJ and Q2W3ER5T6Y7U play them from the computer keyboard.



Every claim in this manual is a check in the instrument's bench, run against the real engine. The numbers below are from build 260904.1, 48 kHz.

| CLAIM          | MEASURED                                                                                                                               |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| reference bowl | a sine, worst harmonic –120 dB, 110.000 Hz                                                                                             |
| isochrone      | a sheared bowl within 0.01 cents from a 0.5 % strike to a full one                                                                     |
| energy         | a frictionless ball keeps its energy to 0.00003 % over a minute                                                                        |
| box bowl       | velocity tap third harmonic –10.6 dB (a square is –9.5); position tap third –20 dB (a triangle is –19)                                 |
| asymmetry      | even harmonics at –20 dB; mirrored, below –120 dB                                                                                      |
| double well    | period 2.6× longer just above the barrier (the saddle dwell); both pits visited above it                                               |
| rock           | sub-octave from ROCK 30 %; a millionth of difference diverges to order one                                                             |
| tide           | a higher ridge takes longer to cross; at TIDE 100 % the ridge is not there; a weak tether at low tide never crosses                    |
| pins           | two identical notes leave byte-identical wakes and audio; HOLD 100 % reaches a pin within a quarter second; a loop repeats within 0.02 |
| aliasing at 4x | non-harmonic floor: parabola –90 dB, box –59 dB, pit –63 dB at B5                                                                      |
| unison         | equal energy: 0.0001 cents of detune; a box at full spread: hundreds of cents, because a box is not isochronous                        |
| detune         | the played ball is 0.000 cents off at any DETUNE; four balls at 40 % swing the envelope 21.9×                                          |
| the starters   | every one within 3 cents of the note at both a brushed and a hammered key, and within 7 dB of each other                               |
| cost           | a four-note chord 2.8 % of one core; 12 voices × 4 balls 27 % at 4x, 19 % at 2x                                                        |

Twenty-nine factory patches, each bounded, exactly silent with no note, deterministic to the sample, and no two alike. 283 checks, all clear.

## Where things live

Patches: `Documents\Brokild patches\High Tide\`. Settings (the remembered patch folder): `%APPDATA%\Brokild\HighTide.settings`. The terrain in a project or a patch is a 16-bit PNG, 512 × 128, four height units over the full range. Plug-in code `HiTd`, company Brokild.

Every control, button, tool and timeline lane carries a hint, and the panel probe refuses to pass if one is missing: 89 of 89.

HIGH TIDE is a Brokild instrument. Free. Windows VST3 and standalone, 64-bit. Design and bench: BrokildApps/HIGH-TIDE-DESIGN.md. Peter Bøggild and Claude, 2026.